Code Review Documentation: Bias Bounty Mapping Equity Challenge, Submission S25

Craig Carlos Ouma*[1]

[1]Independent Researcher

November 4, 2026

Abstract

This document accompanies the code for submission S25 (`2ncgdcQQ`) to the Bias Bounty Mapping Equity Challenge. S25 scored 0.021520001 on the public leaderboard and 0.019332115 on the private leaderboard. The solution computes the transport, building and point-of-interest coverage gaps per Census tract by comparing the Overture Maps release in the challenge package with reference counts and lengths built from public sources, and combines them into the composite score. Every step runs as a Kaggle script kernel driven from the command line. The document follows Zindi's documentation guideline: overview, architecture, extraction and transformation, model, inference, run time, metrics, error handling and maintenance, followed by the environment and the exact steps to reproduce the submitted file.

1 Overview and objectives

The challenge scores each of 9,794 Census tracts in five hazard regions with a coverage gap between 0 (Overture matches the reference) and 1 (Overture holds nothing the reference holds). The score is the mean of three components that are defined for the tract: transport, building and points of interest. No labels are provided. The objective of the code is to produce, for every tract, the 17-column file in the format of the sample submission, with the composite score that the leaderboard evaluates and the component values and flags that feed the organisers' Bias Scorecard.

The solution computes each component directly from its definition. For each tract it measures the Overture side from the challenge package and the reference side from the public sources listed in Section 3.1, applies the gap function and averages the defined components. There is no trained model in S25 and no random draw affects its output.

2 Architecture

Figure 1 shows the data flow. Each box is one Kaggle script kernel. A kernel reads its inputs from the outputs of the kernels it depends on, which Kaggle mounts under `/kaggle/input`, and writes its outputs to `/kaggle/working`. The laptop only edits code, runs unit tests on synthetic data and drives Kaggle through its command line.

*Correspondence: `craigcarlos95@gmail.com`. Zindi: `craigouma`. Reviewed submission: `2ncgdcQQ`, file `S25_20261030-1227-03_model-45a6fcc.csv`. Code: `https://github.com/craigouma/bias-bounty-mapping-equity-challenge`, tag `code-review`.

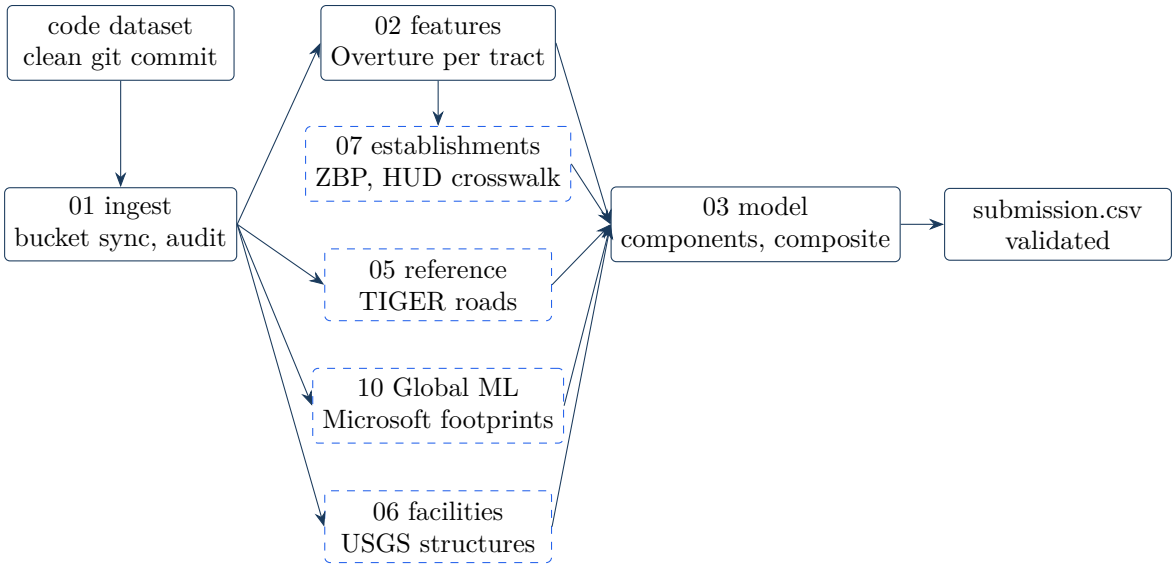


Figure 1: Kaggle stages on the path to S25. Solid boxes read the challenge package only; dashed boxes also download public reference sources from their publishers. Every stage receives the code dataset. Stage 08 (an archived facility layer) runs as an input of stage 03 but is not read by the S25 configuration; stages 00 (environment report), 04 (analysis) and 09 (a diagnostic) are outside the submission path.

3 Extract, transform and load

3.1 Extract

Challenge package. Stage 01 lists the public Source Cooperative bucket (`s3://us-west-2.opendata.source.coop/humane-intelligence/bias-bounty-mapping-equity-challenge/`) without credentials, stops if any file name matches a denylist that includes `coverage-gap`, and downloads 172 files (4,615,029,100 bytes): the six Overture layers of release 2026-08-19.0 per region, the ACS housing table, the per-region sample submissions, the strata tables, the tract polygons and the boundaries. It compares every local byte count with the bucket listing and retries a mismatched file up to three times before failing. The challenge data are read only from these files.

Public reference sources. S25 also uses the sources in Table 1. Each is downloaded inside its kernel from the URL in `configs/data.yaml` at run time, except the HUD crosswalk (see the table note). Retrieval dates are those of the runs that produced S25; the from-scratch rerun in Section 11 downloaded them again on 4 November 2026.

3.2 Transform

Stage 02, features. One row per scored tract. Overture road segments of class motorway, trunk, primary and secondary are clipped to each tract polygon and measured in geodesic metres. Overture footprints are counted by centroid in tract. Overture places are counted per tract in total and by `categories.primary` group (fire department, school types) following the bucket README. Region comes from the `region` column of the sample submission.

Stage 05, reference. TIGER/Line roads with MTFCC S1100 or S1200 are clipped to the tract

Table 1: Public reference sources read by S25. URLs are listed below the table and in `configs/data.yaml`.

	Source	Version	Stage	Retrieved
1	Census TIGER/Line primary and secondary roads	2025	05	27 Oct 2026
2	USGS National Map structures, state GeoPackages	2026	06	28 Oct 2026
3	Census ZIP Code Business Patterns, totals	2023	07	30 Oct 2026
4	HUD USPS ZIP-TRACT crosswalk[a]	Q4 2024	07	29 Oct 2026
5	Census 2020 ZCTA to tract relationship file[b]	2020	07	30 Oct 2026
6	Microsoft Global ML Building Footprints	2026-02-03	10	30 Oct 2026

1. `https://www2.census.gov/geo/tiger/TIGER2025/PRISECROADS/tl_2025_{state}_prisecroads.zip`
2. `https://prd-tnm.s3.amazonaws.com/StagedProducts/Struct/GPKG/STRUCT_{state}_State_GPKG.zip`
3. `https://www2.census.gov/programs-surveys/cbp/datasets/2023/zbp23totals.zip`
4. `https://www.datalumos.org/datalumos/project/219325/version/V2/view`
5. `https://www2.census.gov/geo/docs/maps-data/data/rel2020/zcta520/tab20_zcta520_tract20_natl.txt`
6. `https://minedbuildings.z5.web.core.windows.net/global-buildings/dataset-links.csv`

[a] HUD serves the crosswalk only through a token API. The archived file `ZIP-Tract/ZIP_TRACT_122024.xlsx` was downloaded by hand from DataLumos project 219325, version V2, into `data/hud_crosswalk/`. `scripts/hud_dataset.py` converts it to parquet with ZIP and tract codes kept as text and publishes it as the Kaggle dataset `<user>/bbme-hud-crosswalk`, which stage 07 reads. This is the only manual step. [b] Downloaded by stage 07 for alternative apportionments (by area, population and Overture places); S25 uses the HUD business-address apportionment only.

polygons and measured the same way. A road on a shared tract boundary intersects both tracts and is counted in each.

Stage 06, facilities. USGS structure points with FCODE 74026 (fire stations) and 73002 to 73005 (schools) are counted per tract by point in polygon.

Stage 07, establishments. ZIP Business Patterns 2023 establishment counts are multiplied by the HUD `BUS_RATIO` of each ZIP and tract pair and summed per tract.

Stage 10, Global ML. The quadkey files covering each region are read from the index, and footprints are counted per tract by centroid.

Spatial work uses DuckDB 1.5.4 with its spatial extension. Two of its behaviours fail silently and are guarded in code and tests: transforming from EPSG:4326 without `always_xy` returns infinite coordinates, and `ST_Length_Spheroid` in longitude and latitude order returns NaN. Every length passes through a helper in `src/bbme/db.py` that flips coordinates and asserts a finite result, and a unit test checks it against `pyproj.Geod` to within 0.1 %. GEOID is read as 11-character text everywhere, because an integer cast drops the leading zero of Arizona's state code.

3.3 Load

Each stage writes parquet tables keyed by GEOID and a manifest file to `/kaggle/working`. Kaggle keeps these outputs as a new version of the kernel. A downstream kernel finds its inputs by searching `/kaggle/input` for the upstream manifest name (`FEATURES_MANIFEST.json`, `REFERENCE_MANIFEST.json`, `FACILITY_MANIFEST.json`, `ESTABLISHMENT_MANIFEST.json`, `GLOBALML_MANIFEST.json`),

so the code does not depend on Kaggle's mount paths. Stage 03 joins every table on GEOID with a one-to-one check.

4 Data model

For an Overture quantity O and a reference quantity R in the same tract,

$$\mathrm{gap}(O, R) = \max\left(0,\ 1 - \frac{O}{R}\right), \qquad \text{defined only when } R > 0. \tag{1}$$

Transport T. O is the clipped Overture length of motorway, trunk, primary and secondary roads; R is the clipped TIGER S1100 and S1200 length.

Building B. O is the Overture footprint count and R the Global ML footprint count, both by centroid.

Points of interest P. The facilities half F is the mean of the defined per-type gaps for fire stations and schools (Overture places in the category against USGS points). The establishments half E compares all Overture places with the apportioned establishment count. P is the mean of the defined halves.

The composite is the mean of the defined components,

$$y = \frac{\sum_{c \in \{T,B,P\}} \delta_c \, c}{\sum_c \delta_c}, \tag{2}$$

where δ_c is 1 when component c is defined. An undefined component is written as 0 with its flag `False`. EMS stations are left out of the facilities half; the bucket README states that EMS rarely binds, and adding it made the public score worse.

The configuration is `configs/model.yaml` with the overrides in `configs/final_A.yaml`:

```
transport.source=tiger building.source=microsoft building.count=globalml
poi.source=reference poi.reference_types=[fire,schools] poi.establishment_weight=bus
```

With these settings every scalar that was ever tuned on the leaderboard during the challenge is at its neutral value (scales and levels 1.0, shrinkage 1.0), so S25 carries no leaderboard-tuned number. The leaderboard was used to choose between whole methods, for example which building release and which apportionment to use.

Feature selection and validation. There is no learned model, so there is no feature selection or train and validation split. The rebuilt transport component was checked against two facts the organisers published: the share of tracts with an undefined component per region, which it matches to within one percentage point in all four documented regions, and the range of Overture to TIGER length ratios across regions (0.71 to 1.59), which it reproduces at both ends. The methodology writeup gives the full tables.

5 Inference

Stage 03 is the inference step. It reads the joined table, computes the components and the composite with vectorised pandas and numpy (`src/bbme/model.py`), writes the 17 columns in the sample's

order, row order and `region` values (`src/bbme/submit.py`), and runs the full submission validator before saving. New data (a later Overture release or newer reference files) enter by changing the URLs and release names in `configs/data.yaml` and rerunning stages 01 to 03; no retraining is needed because nothing is trained. Every run is identified by `YYYYMMDD-HHMM-<stage>-<commit>`, and every output folder holds the resolved configuration, the commit hash, `pip freeze`, runtime and peak memory.

6 Run time

Table 2 gives the wall time and peak memory of each stage on Kaggle CPU sessions (4 cores, 32.9 GB RAM), for the runs that produced S25 and for the from-scratch rerun of 4 November 2026. Wall time is measured from push to completion and includes queueing. The full chain takes about 45 minutes when the independent stages run in parallel: stage 01, then stage 05 (the longest of the parallel group), then stage 03.

Table 2: Stage cost on Kaggle CPU sessions.

	Runs behind S25		Rerun, 4 Nov	
Stage	Wall (min)	Peak RSS (GB)	Wall (min)	Peak RSS (GB)
01 ingest	5.4	1.39	7.8	1.47
02 features	27.9	6.99	25.5	6.87
05 reference	33.9	3.55	32.5	3.51
06 facilities	8.7	0.49	26.8	0.50
07 establishments	1.2	0.22	1.1	0.22
08 archive	1.2	0.45	1.2	0.50
10 Global ML	33.8	2.13	16.1	2.22
03 model	1.7	0.56	1.5	0.56

7 Performance metrics

- Public leaderboard RMSE: 0.021520001. Private leaderboard RMSE: 0.019332115.
- Path to S25 on the public leaderboard: all-zero file 0.1072; best estimate from the challenge package alone (S16) 0.0745; transport, building, facility and establishment references rebuilt from public sources (S19) 0.0300; establishments apportioned by HUD business-address ratios (S24) 0.0238; building reference from Global ML (S25) 0.0215.
- Reproduction: a from-scratch rerun gives a file byte-identical to the upload, so the largest absolute difference over every float cell is 0 and no flag differs (Section 11).
- Calibration of the transport component against the organisers' published rates and ratios, described above.

8 Error handling and logging

- **Stale code.** Each kernel starts with a bootstrap that finds the code dataset under `/kaggle/input` and stops if its commit hash differs from the one the kernel was rendered with. The code dataset is only built from a clean git tree.

- **Truncated downloads.** Stage 01 checks every byte count against the bucket listing, retries up to three times and fails the stage on any remaining mismatch.
- **File guard.** Every reader opens files through `src/bbme/guard.py`, which accepts only allowlisted names from `configs/data.yaml` and always rejects names containing `coverage-gap` or `coverage_gap`. A unit test fails if a module opens a file without it.
- **Silent numeric failures.** Length and area helpers assert finite results; GEOIDs are asserted to be 11 digits after every read; joins are checked one to one.
- **Output validation.** Stage 03 refuses to save a file that fails the validator: header and column order, GEOID set, row order and `region` values equal to the sample; finite floats in $[0, 1]$ in plain decimal notation; flags exactly `True` or `False`; a `False` flag with a 0 gap; consistent POI flags; at least one defined component per row.
- **Logs.** Kaggle keeps each kernel's log with its version. The runner saves the log locally when a stage fails, and appends every run (run ID, stage, commit, overrides, state, wall time, peak memory) to `docs/experiments.md`. Every submission, with its run ID, commit, configuration and score, is in `docs/submissions.md`.

9 Maintenance and monitoring

- **Tests.** `make lint test` runs ruff and 178 unit tests on synthetic fixtures, covering GEOID handling, coordinate order, the guard, the gap and composite rules, each feature module and the submission validator.
- **Updating inputs.** Overture release names, bucket paths and reference URLs live in `configs/data.yaml`. A new Overture release or reference vintage needs a rerun of stages 01 to 03 and no code change, unless a schema changes; the stage 01 audit records schemas and distinct values to detect that.
- **Scaling.** Spatial joins run one region at a time with bounding-box pruning before exact intersection, and peak memory stays under 7 GB. More regions add rows linearly.
- **Dependency drift.** `requirements-kaggle.txt` pins every package the stages import. If Kaggle updates its image, the pins restore the versions used for S25.

10 Environment and libraries

- Compute: Kaggle script kernels on CPU sessions, Python 3.13.15, 4 cores, 32.9 GB RAM, internet enabled (the account needs phone verification for internet in kernels). No GPU is used on the S25 path.
- Orchestration: an Intel MacBook Pro with 8 GB RAM running macOS, the Kaggle command-line client, `uv` and GNU Make. Local Python 3.13 with the versions locked in `uv.lock`.
- Libraries installed in every kernel (`requirements-kaggle.txt`; the same list is `requirements.txt` in the submitted archive): duckdb 1.5.4, geopandas 1.1.4, matplotlib 3.10.0, numpy 2.1.3, pandas 2.3.3, pyarrow 23.0.1, pyogrio 0.13.0, pyproj 3.7.2, PyYAML 6.0.3, shapely 2.1.2. All are open source.
- Seed: 20261025 in `configs/model.yaml`, the only source of randomness. The S25 computation makes no random draw.

11 Reproducing S25

The reviewer needs a Kaggle account with internet enabled in kernels, the Kaggle command-line client authenticated on the machine, `git`, `uv` and GNU Make.

1. Clone `https://github.com/craigouma/bias-bounty-mapping-equity-challenge` and check out the tag `code-review`. Run `make setup`.
2. Put the challenge `SampleSubmission.csv` in `data/` and the uploaded S25 file at `artifacts/submissions/S25_20261030-1227-03_model-45a6fcc.csv`.
3. Set `username` in `configs/kaggle.yaml` to the reviewer's Kaggle user name and commit the change; the code dataset is built only from a clean tree.
4. Download `ZIP_TRACT_122024.xlsx` (Table 1, note *a*) into `data/hud_crosswalk/` and run `cdscripts&&uvrunpythonhud_dataset.py`.
5. `make kaggle-code`, then `make kaggle-run STAGE=01_ingest`.
6. Run stages `02_features`, `05_reference`, `06_facilities`, `08_archive` and `10_globalml` (independent of one another), then `07_establishments`, each with `make kaggle-run STAGE=<stage>`.
7. `make reproduce FINAL=A` runs stage 03 with `configs/final_A.yaml` and prints the largest absolute difference and the number of differing flags against the uploaded file.

Each command renders the kernel, pushes it, follows its log, waits for it to finish and downloads its small outputs to `artifacts/kaggle/`. `make kaggle-run` accepts `PREFIX=<text>` to run a second chain of kernels under different names.

Reproduction record. Stage 03 was rerun with the S25 configuration on 30 October 2026 (run `20261030-1512-03_model-8f446f4`) against the upstream outputs that produced S25, and matched the uploaded file in every cell with a largest difference of 0. On 4 November 2026 the whole chain was rerun from scratch under new kernel names (`repro-bbme-*`), downloading the challenge package and every public source again (stage 01 at commit `0f182ec`, stages 02 to 10 at `db9b78d`, stage 07 at `b11465b`, stage 03 at `64296a6`; these commits differ only in run log entries). Every intermediate table read by S25 (TIGER lengths, USGS facility counts, establishment counts, Global ML counts) was identical to the one behind the uploaded file, and the Overture feature table differed only in floating-point summation order (at most 4×10^{-9} m in six columns that S25 does not read). Stage 03 (run `20261104-1651-03_model-64296a6`) produced a file byte-identical to the uploaded S25 file:

SHA-256 `03678f50bac04dea7c2f03abb446deb1fba45b5f55f1a7e716f2d95dbd9ff48d`

The installed versions of every pinned package equalled those of the original S25 run. The commits from `64296a6` to the tag `code-review` change comments, documentation and logs only, which `gitdiff64296a6code-review--srckaggleconfigsrequirements-kaggle.txt` shows.

12 Data insights

Stage 04 holds the exploratory analysis: unnamed road shares by tribal status, rurality and region, maps of the composite score, and the per-stratum tables behind the Best Bias Discovery entry.

The finding, that local streets on federally recognised reservations are unnamed in Overture 9.5 percentage points more often than on matched non-tribal tracts, is written up at `https://craigoguma.github.io/bias-bounty-mapping-equity-challenge/bias-discovery/`.

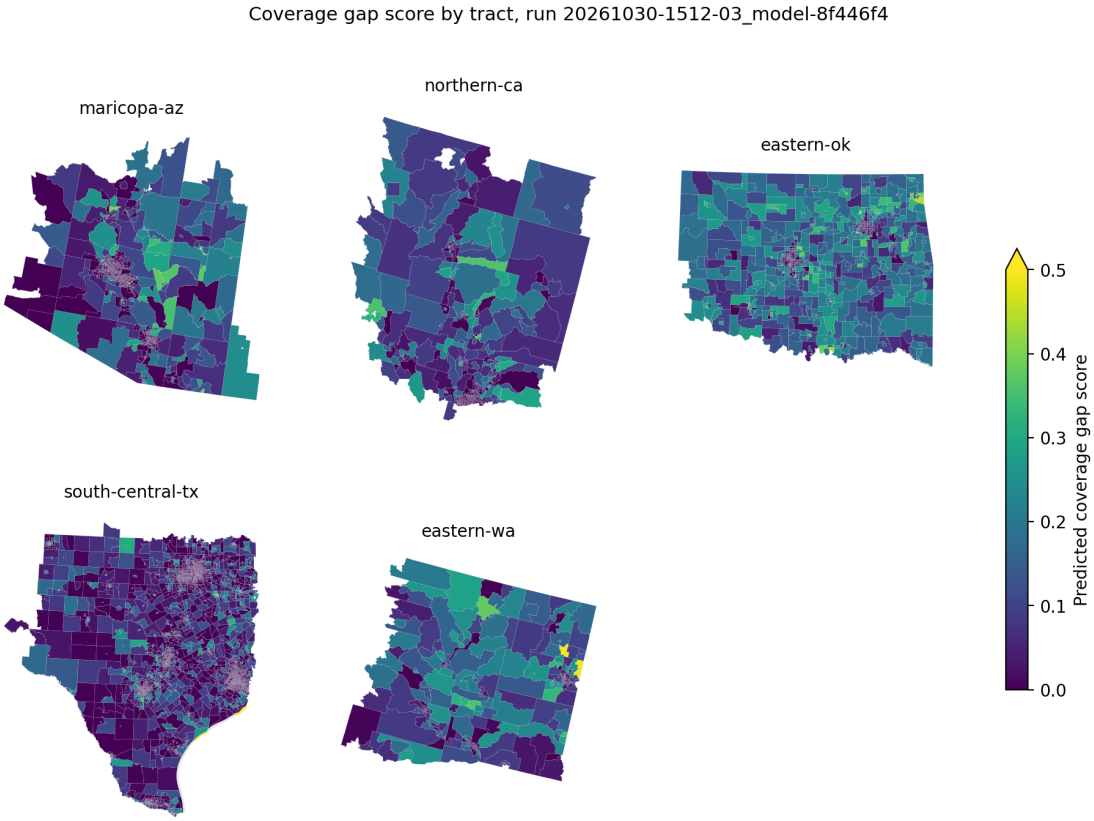


Figure 2: S25 composite coverage gap score per tract.